



NCI Protocol #: 10323

Patient ID:

Date:

1 of 19

Patient Name:
Date Collected:
Biopsy Site: **Colon**
Sample ID:
Telephone:
Tumor Content (%): **20_49%**

Patient DOB:
Referring Physician:
Primary Diagnosis: **Colorectal Cancer**
MoCha ID:
Fax:

Tumor content assessment performed by Van Andel Research Institute; Grand Rapids; MI 49503

Table of Contents

Page

Variant Details	2
Biomarker Descriptions	3
Relevant Therapy Summary	4
Clinical Trials Summary	10

Report Highlights

3 Relevant Biomarkers
30 Therapies Available
60 Clinical Trials

Relevant Colorectal Cancer Findings

Gene	Finding
BRAF	Not detected
KRAS	Not detected
NRAS	Not detected
NTRK1	Not detected
NTRK2	Not detected
NTRK3	Not detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	ERBB2 amplification	lapatinib + trastuzumab pertuzumab + trastuzumab trastuzumab deruxtecan	ado-trastuzumab emtansine ¹ irbinitinib + trastuzumab + chemotherapy ¹ lapatinib + chemotherapy ¹ lapatinib + hormone therapy ¹ margetuximab + chemotherapy ¹ neratinib ¹ neratinib + chemotherapy ¹ pertuzumab + trastuzumab + chemotherapy ¹	41

Public data sources included in relevant therapies: FDA¹, NCCN

Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

* Includes biosimilars

Relevant Biomarkers (continued)

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
			pertuzumab/trastuzumab/ hyaluronidase-zzxf + chemotherapy¹ trastuzumab deruxtecan¹ trastuzumab*¹ trastuzumab* + chemotherapy¹ hormone therapy lapatinib + trastuzumab lapatinib + trastuzumab + hormone therapy pertuzumab + trastuzumab pertuzumab + trastuzumab + hormone therapy + chemotherapy trastuzumab + hormone therapy trastuzumab + hormone therapy + chemotherapy	
IIC	CHEK2 T367fs	None	olaparib ¹	19
IIC	TP53 R174W	None	None	13

Public data sources included in relevant therapies: FDA¹, NCCN

Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

* Includes biosimilars

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
TP53	p.(R174W)	c.520A>T	.	chr17:7578410	11.99%	NM_000546.5	missense
CHEK2	p.(T367fs)	c.1100delC	.	chr22:29091856	58.69%	NM_007194.3	frameshift Deletion

Copy Number Variations

Gene	Locus	Copy Number
ERBB2	chr17:37868167	6.99

Comments:

The CHEK2 T367Mfs variant has been associated with familial cancer syndromes. This assay cannot distinguish between somatic and germline variants. Consultation with clinical genetic service is recommended if clinically prudent.

Signed By: _____

Biomarker Descriptions

CHEK2 (checkpoint kinase 2)

Background: The CHEK2 gene encodes the checkpoint kinase-2 serine/threonine kinase, which is a cell-cycle checkpoint regulator. In response to DNA damage, CHEK2 is phosphorylated by ATM and subsequently phosphorylates and negatively regulates CDC25C to prevent entry into mitosis¹. CHEK2 also stabilizes p53, leading to cell-cycle arrest in G1 phase, and is capable of phosphorylating BRCA1 and promoting DNA repair including homologous recombination repair (HRR)^{2,3,4}. Germline mutations in the CHEK2 gene are associated with Li-Fraumeni syndrome and inherited risk of breast cancer^{5,6,7}.

Alterations and prevalence: Consistent with its role as a tumor suppressor, CHEK2 is enriched for deleterious truncating mutations. Somatic mutations in CHEK2 are common (2-6%) in uterine carcinoma, bladder carcinoma, and lung adenocarcinoma^{8,9}. CHEK2 gene deletions are observed in adrenocortical carcinoma, thymoma, and prostate cancer^{8,9}.

Potential relevance: The PARP inhibitor, olaparib¹⁰ is approved (2020) for metastatic castration-resistant prostate cancer (mCRPC) with deleterious or suspected deleterious, germline or somatic mutations in HRR genes that includes CHEK2.

ERBB2 (erb-b2 receptor tyrosine kinase 2)

Background: The ERBB2 gene encodes the erb-b2 receptor tyrosine kinase 2, a member of the human epidermal growth factor receptor (HER) family. Along with ERBB2/HER2, EGFR/ERBB1/HER1, ERBB3/HER3, and ERBB4/HER4 make up the HER protein family¹¹. All ERBB/HER proteins encode transmembrane receptor tyrosine kinases. However, ERBB2/HER2 is an orphan receptor with no known ligand. ERBB2 preferentially binds other ligand bound ERBB/HER family members to form hetero-dimers resulting in the activation of ERBB2 tyrosine kinase activity and subsequent activation of the PI3K/AKT/MTOR and RAS/RAF/MAPK/ERK signaling pathways which promote cell proliferation, differentiation, and survival¹². Recurrent focal amplification of the ERBB2 gene leads to increased expression in several cancer types. ERBB2 overexpression in immortalized cell lines is oncogenic and leads to ERBB2 homo-dimerization and activation without ligand binding^{13,14,15}.

Alterations and prevalence: ERBB2 gene amplification occurs in 10-20% of breast, esophageal, and gastric cancers, 5-10% of bladder, cervical, pancreas, and uterine cancers, and 1-5% of colorectal, lung, and ovarian cancers^{8,9,16,17,18,19,20,21}. Recurrent somatic activating mutations in ERBB2/HER2 occur at low frequencies (<1%) in diverse cancer types^{9,22,23}. In breast, bladder, and colorectal cancers, the most common recurrent ERBB2 activating mutations include kinase domain mutations L755S and V777L and the extracellular domain mutation S310F. In lung cancer, the most common recurrent ERBB2 activating mutations include in-frame exon 20 insertions, particularly Y772_A775dup.

Potential relevance: The discovery of ERBB2/HER2 as an important driver of breast cancer in 1987 led to the development of trastuzumab, a humanized monoclonal antibody with specificity to the extracellular domain of HER2^{24,25}. Trastuzumab²⁶ was FDA approved for the treatment of HER2 positive breast cancer in 1998, and subsequently in HER2 positive metastatic gastric and gastroesophageal junction adenocarcinoma in 2010. Additional monoclonal antibody therapies have been approved by the FDA for HER2-positive breast cancer including pertuzumab²⁷ (2012), a humanized monoclonal antibody that inhibits HER2 dimerization, and ado-trastuzumab emtansine²⁸ (2013), a conjugate of trastuzumab and a potent antimicrotubule agent. The combination of pertuzumab, trastuzumab, and a taxane is the preferred front-line regimen for HER2-positive metastatic breast cancer²⁹. In addition to monoclonal antibodies, the small molecule inhibitor lapatinib³⁰, with specificity for both EGFR and ERBB2, was FDA approved (2007) for the treatment of patients with advanced HER2-positive breast cancer who have received prior therapy including trastuzumab. In 2017, the FDA approved the use of neratinib³¹, an irreversible kinase inhibitor of EGFR, ERBB2/HER2, and ERBB4, for the extended adjuvant treatment of adult patients with early stage HER2-positive breast cancer. In 2020, the FDA approved neratinib³¹ in combination with capecitabine for HER2-positive advanced or metastatic patients after two or more prior HER2-directed therapies. Also in 2020, the TKI irbitinib³² was FDA approved for HER2 overexpressing or amplified breast cancer in combination with trastuzumab and capecitabine. The vaccine, nelipepimut-S³³, was granted fast-track designation by the FDA (2016) in patients with low to intermediate

Biomarker Descriptions (continued)

HER2 expressing (IHC score 1+ or 2+) breast cancer. In 2018 fast-track designation was granted to the monoclonal antibody margetuximab³⁴ in patients with ERBB2 positive breast cancer previously treated with an anti-HER2 therapy. In 2019, the novel bispecific antibody ZW25³⁵ received fast-track designation for patients with HER2-amplified biliary tract cancer or in combination with standard chemotherapy for patients with HER2-overexpressing gastroesophageal adenocarcinoma (GEA). In 2020, BDTX-189³⁶ received fast-track designation for adult patients with solid tumors harboring an allosteric human ERBB2 mutation or exon 20 insertion, and the humanized anti-HER2 antibody drug conjugate disitamab vedotin received breakthrough designation for adult patients with HER2-positive urothelial cancer after previous platinum-chemotherapy treatment³⁷. In 2021, the antibody-drug conjugate ARX788³⁸ received fast-track designation as a monotherapy for advanced or metastatic HER2-positive breast cancer that have progressed on one or more anti-HER2 regimens. Certain activating mutations have been observed to impart sensitivity to neratinib, afatinib, lapatinib, and trastuzumab, or dacomitinib in early and ongoing clinical studies^{39,40,41,42,43}. ERBB2 kinase domain mutations R896G and V659E both showed response to afatinib in two NSCLC case studies^{44,45}. Additionally, acquired HER2 mutations in estrogen receptor-positive (ER+) breast cancer have been shown to confer resistance to hormone therapy⁴⁶. However, this was shown to be overcome by neratinib in combination with therapies targeting ER⁴⁶.

TP53 (tumor protein p53)

Background: The TP53 gene encodes the p53 tumor suppressor protein that binds to DNA and activates transcription in response to diverse cellular stresses to induce cell cycle arrest, apoptosis, or DNA repair. In unstressed cells, TP53 is kept inactive by targeted degradation via MDM2, a substrate recognition factor for ubiquitin-dependent proteolysis. Alterations in TP53 is required for oncogenesis as they result in loss of protein function and gain of transforming potential⁴⁷. Germline mutations in TP53 are the underlying cause of Li-Fraumeni syndrome, a complex hereditary cancer predisposition disorder associated with early-onset cancers^{48,49}.

Alterations and prevalence: TP53 is the most frequently mutated gene in the cancer genome with approximately half of all cancers experiencing TP53 mutations. Ovarian, head and neck, esophageal, and lung squamous cancers have particularly high TP53 mutation rates (60-90%)^{8,9,50,51,52,53}. Approximately two-thirds of TP53 mutations are missense mutations and several recurrent missense mutations are common including substitutions at codons R158, R175, Y220, R248, R273, and R282^{8,9}. Invariably, recurrent missense mutations in TP53 inactivate its ability to bind DNA and activate transcription of target genes^{54,55,56,57}.

Potential relevance: The small molecule p53 reactivator, PC14586, received a fast track designation (2020) by the FDA for advanced tumors harboring a TP53 Y220C mutation⁵⁸. The FDA has granted fast track designation (2019) to the p53 reactivator, APR-246 alone⁵⁹ and breakthrough designation⁶⁰ (2020) in combination with azacitidine for myelodysplastic syndrome (MDS) and acute myeloid leukemia patients (AML) harboring a TP53 mutation. In addition to investigational therapies aimed at restoring wild-type TP53 activity, compounds that induce synthetic lethality are also under clinical evaluation^{61,62}. TP53 mutations confer poor prognosis in multiple blood cancers including AML, MDS, myeloproliferative neoplasms (MPN), chronic lymphocytic leukemia (CLL)^{63,64,65,66}. In mantle cell lymphoma, TP53 mutations are associated with poor prognosis when treated with conventional therapy including hematopoietic cell transplant⁶⁷. Mono- and bi-allelic mutations in TP53 confer unique characteristics in MDS, with multi-hit patients also experiencing associations with complex karyotype, few co-occurring mutations, and high-risk disease presentation as well as predicted death and leukemic transformation independent of the IPSS-R staging system⁶⁸.

Relevant Therapy Summary

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ERBB2 amplification

Relevant Therapy	FDA	NCCN	Clinical Trials*
trastuzumab deruxtecan	☐	☒	☒

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ERBB2 amplification (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
ado-trastuzumab emtansine	☐	☐	☒ (II)
trastuzumab	☐	☐	☒ (II)
irbinitinib + trastuzumab + capecitabine	☐	☐	☒
lapatinib + capecitabine	☐	☐	☒
neratinib	☐	☐	☒
neratinib + capecitabine	☐	☐	☒
pertuzumab + trastuzumab + chemotherapy	☐	☐	☒
pertuzumab + trastuzumab + docetaxel	☐	☐	☒
trastuzumab + capecitabine + cisplatin	☐	☐	☒
trastuzumab + carboplatin + docetaxel	☐	☐	☒
trastuzumab + cisplatin + fluorouracil	☐	☐	☒
trastuzumab + docetaxel	☐	☐	☒
trastuzumab + paclitaxel	☐	☐	☒
lapatinib + letrozole	☐	☒	☒
margetuximab + chemotherapy	☐	☒	☒
pertuzumab/trastuzumab/hyaluronidase-zzxf + cyclophosphamide + doxorubicin	☐	☒	☒
pertuzumab/trastuzumab/hyaluronidase-zzxf + docetaxel	☐	☒	☒
trastuzumab (Biocon)	☐	☒	☒
trastuzumab (Biocon) + capecitabine + cisplatin	☐	☒	☒
trastuzumab (Biocon) + carboplatin + docetaxel	☐	☒	☒
trastuzumab (Biocon) + cisplatin + fluorouracil	☐	☒	☒
trastuzumab (Biocon) + docetaxel	☐	☒	☒
trastuzumab (Biocon) + paclitaxel	☐	☒	☒
trastuzumab (Celltrion)	☐	☒	☒
trastuzumab (Celltrion) + capecitabine + cisplatin	☐	☒	☒
trastuzumab (Celltrion) + carboplatin + docetaxel	☐	☒	☒

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ERBB2 amplification (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
trastuzumab (Celltrion) + cisplatin + fluorouracil	☐	×	×
trastuzumab (Celltrion) + docetaxel	☐	×	×
trastuzumab (Celltrion) + paclitaxel	☐	×	×
trastuzumab (Enhance)	☐	×	×
trastuzumab (Enhance) + carboplatin + docetaxel	☐	×	×
trastuzumab (Enhance) + docetaxel	☐	×	×
trastuzumab (Enhance) + paclitaxel	☐	×	×
trastuzumab (Pfizer)	☐	×	×
trastuzumab (Pfizer) + capecitabine + cisplatin	☐	×	×
trastuzumab (Pfizer) + carboplatin + docetaxel	☐	×	×
trastuzumab (Pfizer) + cisplatin + fluorouracil	☐	×	×
trastuzumab (Pfizer) + docetaxel	☐	×	×
trastuzumab (Pfizer) + paclitaxel	☐	×	×
trastuzumab (Samsung Bioepis)	☐	×	×
trastuzumab (Samsung Bioepis) + capecitabine + cisplatin	☐	×	×
trastuzumab (Samsung Bioepis) + carboplatin + docetaxel	☐	×	×
trastuzumab (Samsung Bioepis) + cisplatin + fluorouracil	☐	×	×
trastuzumab (Samsung Bioepis) + docetaxel	☐	×	×
trastuzumab (Samsung Bioepis) + paclitaxel	☐	×	×
pertuzumab + trastuzumab	×	☒	☒ (II)
lapatinib + trastuzumab	×	☒	×
aromatase inhibitor	×	☐	×
fulvestrant	×	☐	×
hormone therapy	×	☐	×
lapatinib + aromatase inhibitor	×	☐	×
lapatinib + trastuzumab + aromatase inhibitor	×	☐	×

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ERBB2 amplification (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
neratinib + paclitaxel	×	○	×
pertuzumab + trastuzumab + carboplatin + docetaxel	×	○	×
pertuzumab + trastuzumab + hormone therapy + chemotherapy	×	○	×
pertuzumab + trastuzumab + paclitaxel	×	○	×
tamoxifen	×	○	×
trastuzumab + aromatase inhibitor	×	○	×
trastuzumab + capecitabine	×	○	×
trastuzumab + capecitabine + oxaliplatin	×	○	×
trastuzumab + carboplatin + docetaxel + fluorouracil	×	○	×
trastuzumab + carboplatin + paclitaxel	×	○	×
trastuzumab + chemotherapy	×	○	×
trastuzumab + chemotherapy (other)	×	○	×
trastuzumab + cisplatin + docetaxel	×	○	×
trastuzumab + cisplatin + docetaxel + fluorouracil	×	○	×
trastuzumab + cisplatin + paclitaxel	×	○	×
trastuzumab + cyclophosphamide + docetaxel	×	○	×
trastuzumab + docetaxel + fluorouracil + oxaliplatin	×	○	×
trastuzumab + fluorouracil	×	○	×
trastuzumab + fluorouracil + irinotecan	×	○	×
trastuzumab + fluorouracil + oxaliplatin	×	○	×
trastuzumab + fulvestrant	×	○	×
trastuzumab + hormone therapy + chemotherapy	×	○	×
trastuzumab + tamoxifen	×	○	×
trastuzumab + vinorelbine	×	○	×
ado-trastuzumab emtansine + atezolizumab	×	×	● (II)
irbinitinib, trastuzumab	×	×	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ERBB2 amplification (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
neratinib + trastuzumab, cetuximab + neratinib	×	×	● (II)
pertuzumab, cetuximab, trastuzumab, chemotherapy	×	×	● (II)
pyrotinib, trastuzumab	×	×	● (II)
trastuzumab, chemotherapy	×	×	● (II)
trastuzumab, pertuzumab, ado-trastuzumab emtansine, lapatinib	×	×	● (II)
A-166	×	×	● (I/II)
BDC-1001, pembrolizumab	×	×	● (I/II)
BDTX-189	×	×	● (I/II)
CART-HER2 (Bellicum Pharmaceuticals), rimiducid	×	×	● (I/II)
zotatifin	×	×	● (I/II)
ado-trastuzumab (Shanghai Fosun Pharma)	×	×	● (I)
AMX-3009	×	×	● (I)
ARX-788	×	×	● (I)
BAY-2701439	×	×	● (I)
CART	×	×	● (I)
CART-HER2	×	×	● (I)
CART-HER2, chemotherapy	×	×	● (I)
irbinitinib, trastuzumab, chemotherapy	×	×	● (I)
KN026	×	×	● (I)
M802	×	×	● (I)
MBS301	×	×	● (I)
MT-5111	×	×	● (I)
neratinib, palbociclib, everolimus, trametinib	×	×	● (I)
pirotinib	×	×	● (I)
pivotinib	×	×	● (I)
PRS-343	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ERBB2 amplification (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
SBT6050, pembrolizumab	×	×	● (I)
SHR-A1811	×	×	● (I)
tebotelimab, margetuximab	×	×	● (I)
zanidatamab	×	×	● (I)
ZN-A-1041	×	×	● (I)

CHEK2 T367fs

Relevant Therapy	FDA	NCCN	Clinical Trials*
olaparib	○	○	● (II)
atezolizumab	×	×	● (II)
niraparib	×	×	● (II)
olaparib, pembrolizumab	×	×	● (II)
rucaparib, atezolizumab	×	×	● (II)
talazoparib	×	×	● (II)
BAY-1895344	×	×	● (I/II)
xevinapant, nivolumab	×	×	● (I/II)
BAY-1895344, niraparib	×	×	● (I)
BAY-1895344, pembrolizumab	×	×	● (I)
copanlisib, olaparib, durvalumab	×	×	● (I)
HH-CYH33, olaparib	×	×	● (I)
HWH-340	×	×	● (I)
talazoparib, palbociclib, axitinib, crizotinib	×	×	● (I)

TP53 R174W

Relevant Therapy	FDA	NCCN	Clinical Trials*
atezolizumab	×	×	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

TP53 R174W (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
niraparib	×	×	● (II)
olaparib	×	×	● (II)
talazoparib	×	×	● (II)
BAY-1895344	×	×	● (I/II)
eprenetapopt, pembrolizumab	×	×	● (I/II)
xevinapant, nivolumab	×	×	● (I/II)
BAY-1895344, niraparib	×	×	● (I)
BAY-1895344, pembrolizumab	×	×	● (I)
HWH-340	×	×	● (I)
talazoparib, palbociclib, axitinib, crizotinib	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

ERBB2 amplification

NCT ID	Title	Phase
NCT03457896	A Phase II Study Evaluating the Combination of Neratinib Plus Trastuzumab or Neratinib Plus Cetuximab in Patients With "Quadruple Wild-Type" (KRAS/NRAS/BRAF/PIK3CA Wild-Type) Metastatic Colorectal Cancer Based on HER2 Status: Amplified, Non-Amplified (Wild-Type) or Mutated	II
NCT04380012	Pyrotinib Maleate With or Without Trastuzumab in the Treatment of HER2-positive Advanced Colorectal Cancer: a Multicenter Clinical Trial	II
NCT03185988	Multicenter, Phase II Study of Chemotherapy in Combination With Trastuzumab in Patients of Pretreated, HER2 Positive, Relapse or Metastatic Carcinoma of Digestive System	II
NCT03602079	A Phase I-II, FIH Study of A166 in Locally Advanced/Metastatic Solid Tumors Expressing Human Epidermal Growth Factor Receptor 2 (HER2) or Are HER2 Amplified That Did Not Respond or Stopped Responding to Approved Therapies	I/II
NCT04650451	A Phase I/II, Open-Label, Multicenter, Non-Randomized, Safety and Activity Study of HER2-Targeted Dual Switch CAR-T Cells (BPX-603) In Subjects With Previously Treated Advanced HER2-Positive Solid Tumors	I/II
NCT04430738	A Phase Ib/II Dose Escalation and Expansion Study of Tucatinib in Combination With Trastuzumab and Oxaliplatin-based Chemotherapy for HER2+ Gastrointestinal Cancers	I

Clinical Trials Summary (continued)

ERBB2 amplification (continued)

NCT ID	Title	Phase
No NCT ID	Phase I Clinical Study With Advanced Solid Tumors KBP-5209 Treatment	I
NCT04460456	A Phase 1/1B, Open-Label, Dose Escalation and Expansion Study of SBT6050 Alone and in Combination With Pembrolizumab in Subjects With Advanced Solid Tumors Expressing HER2	I
NCT02892123	Phase I Trial of ZW25 in Patients With Locally Advanced (Unresectable) and/or Metastatic HER2-expressing Cancers.	I
No NCT ID	TR Study of "Multicenter Phase II Study to Evaluate Efficacy and Safety of Combination Therapy with Trastuzumab and Pertuzumab in Patients with HER2-Positive Metastatic Colorectal Cancer"	II
NCT03843749	Pyrotinib in Combination With Trastuzumab in Treatment-Refractory, HER2-Positive Metastatic Colorectal Cancer.	II
NCT02675829	A Phase II Trial of Ado-Trastuzumab Emtansine for Patients With HER2 Amplified or Mutant Cancers	II
NCT03418558	Open-label, Phase II Study of Trastuzumab Emtansine in Patients With HER2-positive Metastatic Colorectal Cancer Progressing After Trastuzumab and Lapatinib: HERACLES RESCUE. (HER2 Amplification for Colo-rectal Cancer Enhanced Stratification - REchallenge With her2 Selective Cytotoxic Uptake of Emtansine)	II
NCT04632992	MyTACTIC: An Open-Label Phase II Study Evaluating Targeted Therapies in Patients Who Have Advanced Solid Tumors With Genomic Alterations or Protein Expression Patterns Predictive of Response	II
NCT03043313	MOUNTAINEER: A Phase II, Open Label Study of Tucatinib Combined With Trastuzumab in Patients With HER2+ Metastatic Colorectal Cancer.	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study	II
NCT03365882	S1613, A Randomized Phase II Study of Trastuzumab and Pertuzumab (TP) Compared to Cetuximab and Irinotecan (CETIRI) in Advanced/Metastatic Colorectal Cancer (mCRC) With HER-2 Amplification	II
NCT03239015	Efficacy and Safety of Targeted Precision Therapy in Refractory Tumor With Druggable Molecular Event	II
NCT04591431	The Rome Trial From Histology to Target: the Road to Personalize Target Therapy and Immunotherapy	II
NCT04278144	Phase I/II Study of BDC-1001 as a Single Agent and in Combination With Pembrolizumab in Patients With Advanced HER2-Expressing Solid Tumors	I/II
NCT04209465	MasterKey-01: A Phase I/II, Open-label, Two-part, Multicenter Study to Assess the Safety, Tolerability, Pharmacokinetics & Antitumor Activity of BDTX-189, an Inhibitor of Allosteric ErbB Mutations, in Patients w/ Advanced Solid Malignancies	I/II
NCT04092673	A Phase 1-2 Dose-Escalation and Cohort-Expansion Study of Intravenous Zotatfin (eFT226) in Subjects With Selected Advanced Solid Tumor Malignancies	I/II
No NCT ID	Phase I Clinical Study of Safety, Tolerability, Pharmacokinetics and Initial Efficacy of A166 in the Treatment of Patients with Locally Advanced or Metastatic Solid Tumors with HER2.	I
NCT03944499	A Phase I, Multicenter, Open-label, Single-arm Study: A Dose-escalation Phase (Phase 1a) Evaluating FS-1502 in Patients With HER2 Expressed Advanced Solid Tumors; and a Dose-expanded Cohort (Phase 1b) Evaluating FS-1502 in Patients With Local Advanced or Metastatic, HER2 Positive Breast Cancer.	I

Clinical Trials Summary (continued)

ERBB2 amplification (continued)

NCT ID	Title	Phase
No NCT ID	Single-Arm, Open-Label, Single-Dose And Multiple-Dose Phase Ia Clinical Study Of Tolerability And Pharmacokinetics Of AMX3009 In Patients With HER2-Positive Advanced Solid Tumors	I
NCT03255070	A Phase I, Multicenter, Open-label, Multiple Dose-escalation Study of ARX788, Intravenously Administered as a Single Agent in Subjects With Advanced Cancers With HER2 Expression	I
NCT04147819	A Phase I Open-label, First-in-human, Multi-center Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Anti-tumor Activity of Thorium-227 Labeled Antibody-chelator Conjugate BAY2701439, in Participants With Advanced HER2-expressing Tumors	I
NCT04511871	A Phase I Trial to Assess Safety, Tolerability and Anti-tumor Activity of Autologous T Cell Modified Chimeric Antigen Receptor (CAR) (CCT303-406) in Patients With Relapsed or Refractory HER2 Positive Solid Tumors	I
NCT03696030	A Phase I Cellular Immunotherapy Study of Intraventricularly Administered Autologous HER2-Targeted Chimeric Antigen Receptor (HER2-CAR) T Cells in Patients With Brain and/or Leptomeningeal Metastases From HER2 Positive Cancers	I
NCT03847168	Phase I Study of KN026 in HER2 Expressing Breast Cancer, Astic/Gastroesophageal Junction Cancer and Other Locally Advanced/Metastatic Solid Tumors	I
NCT04501770	A Phase I Study to Evaluate the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics and Immunogenicity Profiles of the Recombinant Anti-HER2 and Anti-CD3 Humanized Bispecific Antibody (M802) in HER2-Positive Advanced Solid Tumors	I
NCT03842085	Evaluation on the Safety and Pharmacokinetics of Recombinant Humanized Bispecific Monoclonal Antibody MBS301 for Injection in Treatment of HER2 Positive Recurrent or Metastatic Malignant Solid Tumor	I
NCT04029922	A Phase I Open-label, Multicenter Dose Escalation Study of MT-5111 in Subjects With Previously Treated Advanced HER2-positive Solid Tumors	I
NCT03065387	Phase I Study of the Pan-ERBB Inhibitor Neratinib Given in Combination With Everolimus, Palbociclib, or Trametinib in Advanced Cancer Subjects With EGFR Mutation/Amplification, HER2 Mutation/Amplification, or HER3/4 Mutation or KRAS Mutation	I
No NCT ID	Tolerance And Pharmacokinetics Of Pivotinib In Patients With Advanced Solid Tumors With HER2 Expression	I
NCT03219268	A Phase I, First-in-Human, Open-Label, Dose Escalation Study of MGD013, A Bispecific DART Protein Binding PD-1 and LAG-3 in Patients With Unresectable or Metastatic Neoplasms	I
NCT04487236	A Phase I Clinical Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Efficacy of ZN-A-1041 Enteric Capsules as a Single Agent or in Combination With Capecitabine Tablets in Patients With HER2-Positive Advanced Solid Tumors	I
NCT02442297	Phase I Study of Intracranial Injection of T Cells Expressing HER2-specific Chimeric Antigen Receptors (CAR) in Subjects With HER2-Positive Tumors of the Central Nervous System (iCAR)	I
NCT03330561	A Phase I, Open-Label, Dose Escalation Study of PRS-343 in Patients With HER2-Positive Advanced or Metastatic Solid Tumors	I

Clinical Trials Summary (continued)

ERBB2 amplification (continued)

NCT ID	Title	Phase
NCT04446260	A Phase I Multi-Country, Multi-Center, Open-Label Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Efficacy of SHR-A1811 in HER2 Expressing or Mutated Advanced Malignant Solid Tumor Subjects	I
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II

CHEK2 T367fs

NCT ID	Title	Phase
NCT04122625	A Dose-optimization, Exploratory Phase Ib/II Study to Assess Safety and Efficacy of the Second Mitochondrial-derived Activator of Caspases (SMAC) Mimetic Debio 1143, When Given in Combination With the Anti-PD-1 Antibody Nivolumab in Patients With Specific Solid Tumors Who Have Progressed During or Immediately After Anti-PD-1/PD-L1 Treatment	I/II
NCT04095273	A Multicenter, Non-randomized, Open-label Phase Ib Study to Determine the Maximum Tolerated and Recommended Phase II Dose of the ATR Inhibitor BAY1895344 in Combination With Pembrolizumab and to Characterize Its Safety, Tolerability, Pharmacokinetics and Preliminary Anti-tumor Activity in Patients With Advanced Solid Tumors	I
NCT03188965	An Open-label, First-in-human, Dose-escalation Study to Evaluate the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics, and Maximum Tolerated Dose and / or Recommended Phase II Dose of the ATR Inhibitor BAY1895344 in Patients With Advanced Solid Tumors and Lymphomas	I/II
NCT04550494	A Pharmacodynamics-Driven Trial of Talazoparib, an Oral PARP Inhibitor, in Patients With Advanced Solid Tumors and Aberrations in Genes Involved in DNA Damage Response	II
NCT03207347	A Phase II Trial of the PARP Inhibitor, Niraparib, in BAP1 and Other DNA Damage Response (DDR) Pathway Deficient Neoplasms (UF-STO-ETI-001).	II
NCT03742895	A Phase II Study of Olaparib Monotherapy in Participants With Previously Treated, Homologous Recombination Repair Mutation (HRRm) or Homologous Recombination Deficiency (HRD) Positive Advanced Cancer	II
NCT03967938	Efficacy of Olaparib in Advanced Cancers Occurring in Patients With Germline Mutations or Somatic Tumor Mutations in Homologous Recombination Genes	II
NCT04123366	A Phase II Study of Olaparib in Combination With Pembrolizumab in Participants With Previously Treated, Homologous Recombination Repair Mutation (HRRm) and/or Homologous Recombination Deficiency (HRD)-Positive Advanced Cancer	II
NCT04276376	A Multicenter, Open Label, Phase II Basket Trial Exploring the Efficacy And Safety of The Combination of Rucaparib (PARP Inhibitor) And Atezolizumab (Anti-PD-L1 Antibody) In Patients With DNA Repair-Deficient or Platinum-Sensitive Solid Tumors	II
NCT02401347	A Phase II Clinical Trial of the PARP Inhibitor Talazoparib in BRCA1 and BRCA2 Wild Type Patients With Advanced Triple Negative Breast Cancer and Homologous Recombination Deficiency or Advanced HER2 Negative Breast Cancer or Other Solid Tumors With a Mutation in Homologous Recombination Pathway Genes Talazoparib Beyond BRCA (TBB) Trial	II
NCT03842228	A Phase Ib Biomarker-Driven Combination Trial of Copanlisib, Olaparib, and MEDI4736 (Durvalumab) in Patients With Advanced Solid Tumors	I

Clinical Trials Summary (continued)

CHEK2 T367fs (continued)

NCT ID	Title	Phase
NCT04586335	Open Label, Phase Ib Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Clinical Activity of CYH33, an Oral α -specific PI3K Inhibitor in Combination With Olaparib, an Oral PARP Inhibitor in Patients With Advanced Solid Tumors.	I
NCT02029001	A Two-period, Multicenter, Randomized, Open-label, Phase II Study Evaluating the Clinical Benefit of a Maintenance Treatment Targeting Tumor Molecular Alterations in Patients With Progressive Locally-advanced or Metastatic Solid Tumors MOST: My own specific treatment	II
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT03415659	A Phase I, Open-label, Single-center, Single/Multiple-dose, Dose-escalation/Dose-expansion Clinical Study on Tolerance and Pharmacokinetics of HWH340 Tablet in Patients With Advanced Solid Tumors	I
NCT03767075	Basket of Baskets: A Modular, Open-label, Phase II, Multicentre Study To Evaluate Targeted Agents in Molecularly Selected Populations With Advanced Solid Tumours	II
NCT03233204	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase 2 Subprotocol of Olaparib in Patients With Tumors Harboring Defects in DNA Damage Repair Genes	II
NCT04267939	An Open-label Phase Ib Study to Determine the Maximum Tolerated and/or Recommended Phase II Dose of the ATR Inhibitor BAY 1895344 in Combination With PARP Inhibitor Niraparib, in Patients With Recurrent Advanced Solid Tumors and Ovarian Cancer	I
NCT04693468	Modular Phase IB Hypothesis-Testing, Biomarker-Driven, Talazoparib Combination Trial (TalaCom)	I

TP53 R174W

NCT ID	Title	Phase
NCT04122625	A Dose-optimization, Exploratory Phase Ib/II Study to Assess Safety and Efficacy of the Second Mitochondrial-derived Activator of Caspases (SMAC) Mimetic Debio 1143, When Given in Combination With the Anti-PD-1 Antibody Nivolumab in Patients With Specific Solid Tumors Who Have Progressed During or Immediately After Anti-PD-1/PD-L1 Treatment	I/II
NCT04095273	A Multicenter, Non-randomized, Open-label Phase Ib Study to Determine the Maximum Tolerated and Recommended Phase II Dose of the ATR Inhibitor BAY1895344 in Combination With Pembrolizumab and to Characterize Its Safety, Tolerability, Pharmacokinetics and Preliminary Anti-tumor Activity in Patients With Advanced Solid Tumors	I
NCT03188965	An Open-label, First-in-human, Dose-escalation Study to Evaluate the Safety, Tolerability, Pharmacokinetics, Pharmacodynamics, and Maximum Tolerated Dose and / or Recommended Phase II Dose of the ATR Inhibitor BAY1895344 in Patients With Advanced Solid Tumors and Lymphomas	I/II
NCT04383938	Study of APR-246 in Combination With Pembrolizumab in Subjects With Solid Tumor Malignancies	I/II
NCT02029001	A Two-period, Multicenter, Randomized, Open-label, Phase II Study Evaluating the Clinical Benefit of a Maintenance Treatment Targeting Tumor Molecular Alterations in Patients With Progressive Locally-advanced or Metastatic Solid Tumors MOST: My own specific treatment	II
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II

Clinical Trials Summary (continued)

TP53 R174W (continued)

NCT ID	Title	Phase
NCT02401347	A Phase II Clinical Trial of the PARP Inhibitor Talazoparib in BRCA1 and BRCA2 Wild Type Patients With Advanced Triple Negative Breast Cancer and Homologous Recombination Deficiency or Advanced HER2 Negative Breast Cancer or Other Solid Tumors With a Mutation in Homologous Recombination Pathway Genes Talazoparib Beyond BRCA (TBB) Trial	II
NCT03415659	A Phase I, Open-label, Single-center, Single/Multiple-dose, Dose-escalation/Dose-expansion Clinical Study on Tolerance and Pharmacokinetics of HWH340 Tablet in Patients With Advanced Solid Tumors	I
NCT03767075	Basket of Baskets: A Modular, Open-label, Phase II, Multicentre Study To Evaluate Targeted Agents in Molecularly Selected Populations With Advanced Solid Tumours	II
NCT03207347	A Phase II Trial of the PARP Inhibitor, Niraparib, in BAP1 and Other DNA Damage Response (DDR) Pathway Deficient Neoplasms (UF-STO-ETI-001).	II
NCT03233204	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase 2 Subprotocol of Olaparib in Patients With Tumors Harboring Defects in DNA Damage Repair Genes	II
NCT04267939	An Open-label Phase Ib Study to Determine the Maximum Tolerated and/or Recommended Phase II Dose of the ATR Inhibitor BAY 1895344 in Combination With PARP Inhibitor Niraparib, in Patients With Recurrent Advanced Solid Tumors and Ovarian Cancer	I
NCT04693468	Modular Phase IB Hypothesis-Testing, Biomarker-Driven, Talazoparib Combination Trial (TalaCom)	I

Assay Information

Methodology, Scope, and Application: The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity: The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations: This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent OncoPrint Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OncoPrint Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trial, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET, MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGF, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSP02, RSP03, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

References

1. Matsuoka et al. Linkage of ATM to cell cycle regulation by the Chk2 protein kinase. *Science*. 1998 Dec 4;282(5395):1893-7. PMID: 9836640
2. Cai et al. Structure and activation mechanism of the CHK2 DNA damage checkpoint kinase. *Mol. Cell*. 2009 Sep 24;35(6):818-29. PMID: 19782031
3. Zhang et al. Chk2 phosphorylation of BRCA1 regulates DNA double-strand break repair. *Mol. Cell. Biol*. 2004 Jan;24(2):708-18. PMID: 14701743
4. Huang et al. Chk1 and Chk2 are differentially involved in homologous recombination repair and cell cycle arrest in response to DNA double-strand breaks induced by camptothecins. *Mol. Cancer Ther*. 2008 Jun;7(6):1440-9. PMID: 18566216
5. Apostolou et al. Current perspectives on CHEK2 mutations in breast cancer. *Breast Cancer (Dove Med Press)*. 2017;9:331-335. PMID: 28553140
6. Nevanlinna et al. The CHEK2 gene and inherited breast cancer susceptibility. *Oncogene*. 2006 Sep 25;25(43):5912-9. PMID: 16998506
7. Näslund-Koch et al. Increased Risk for Other Cancers in Addition to Breast Cancer for CHEK2*1100delC Heterozygotes Estimated From the Copenhagen General Population Study. *J. Clin. Oncol*. 2016 Apr 10;34(11):1208-16. PMID: 26884562
8. Weinstein et al. The Cancer Genome Atlas Pan-Cancer analysis project. *Nat. Genet*. 2013 Oct;45(10):1113-20. PMID: 24071849
9. Cerami et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov*. 2012 May;2(5):401-4. PMID: 22588877
10. https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/208558s019s020lbl.pdf
11. King et al. Amplification of a novel v-erbB-related gene in a human mammary carcinoma. *Science*. 1985 Sep 6;229(4717):974-6. PMID: 2992089
12. Gutierrez et al. HER2: biology, detection, and clinical implications. *Arch. Pathol. Lab. Med*. 2011 Jan;135(1):55-62. PMID: 21204711
13. Di et al. erbB-2 is a potent oncogene when overexpressed in NIH/3T3 cells. *Science*. 1987 Jul 10;237(4811):178-82. PMID: 2885917
14. Hudziak et al. Increased expression of the putative growth factor receptor p185HER2 causes transformation and tumorigenesis of NIH 3T3 cells. *Proc. Natl. Acad. Sci. U.S.A.* 1987 Oct;84(20):7159-63. PMID: 2890160
15. Lonardo et al. The normal erbB-2 product is an atypical receptor-like tyrosine kinase with constitutive activity in the absence of ligand. *New Biol*. 1990 Nov;2(11):992-1003. PMID: 1983208
16. Ciriello et al. Comprehensive Molecular Portraits of Invasive Lobular Breast Cancer. *Cell*. 2015 Oct 8;163(2):506-19. PMID: 26451490
17. Cancer Genome Atlas Research Network. Comprehensive molecular characterization of gastric adenocarcinoma. *Nature*. 2014 Sep 11;513(7517):202-9. doi: 10.1038/nature13480. Epub 2014 Jul 23. PMID: 25079317
18. Cancer Genome Atlas Research Network. Comprehensive molecular characterization of urothelial bladder carcinoma. *Nature*. 2014 Mar 20;507(7492):315-22. doi: 10.1038/nature12965. Epub 2014 Jan 29. PMID: 24476821
19. Cancer Genome Atlas Network. Comprehensive molecular characterization of human colon and rectal cancer. *Nature*. 2012 Jul 18;487(7407):330-7. PMID: 22810696
20. Cancer Genome Atlas Research Network. Comprehensive molecular profiling of lung adenocarcinoma. *Nature*. 2014 Jul 31;511(7511):543-50. doi: 10.1038/nature13385. Epub 2014 Jul 9. PMID: 25079552
21. Cancer Genome Atlas Research Network. Integrated genomic analyses of ovarian carcinoma. *Nature*. 2011 Jun 29;474(7353):609-15. PMID: 21720365
22. Petrelli et al. Clinical and pathological characterization of HER2 mutations in human breast cancer: a systematic review of the literature. *Breast Cancer Res. Treat*. 2017 Nov;166(2):339-349. PMID: 28762010
23. Bose et al. Activating HER2 mutations in HER2 gene amplification negative breast cancer. *Cancer Discov*. 2013 Feb;3(2):224-37. doi: 10.1158/2159-8290.CD-12-0349. Epub 2012 Dec 7. PMID: 23220880
24. Hudis. Trastuzumab—mechanism of action and use in clinical practice. *N. Engl. J. Med*. 2007 Jul 5;357(1):39-51. PMID: 17611206
25. Slamon et al. Human breast cancer: correlation of relapse and survival with amplification of the HER-2/neu oncogene. *Science*. 1987 Jan 9;235(4785):177-82. PMID: 3798106

References (continued)

26. https://www.accessdata.fda.gov/drugsatfda_docs/label/2018/103792s5345lbl.pdf
27. https://www.accessdata.fda.gov/drugsatfda_docs/label/2020/125409s124lbl.pdf
28. https://www.accessdata.fda.gov/drugsatfda_docs/label/2020/125427s108lbl.pdf
29. NCCN Guidelines® - NCCN-Breast Cancer [Version 1.2021]
30. https://www.accessdata.fda.gov/drugsatfda_docs/label/2018/022059s024lbl.pdf
31. https://www.accessdata.fda.gov/drugsatfda_docs/label/2020/208051s007lbl.pdf
32. https://www.accessdata.fda.gov/drugsatfda_docs/label/2020/213411s000lbl.pdf
33. <https://www.globenewswire.com/news-release/2016/06/01/845166/0/en/Galena-Biopharma-Receives-Fast-Track-Designation-for-NeuVax-nelipepimut-S-PRESENT-Clinical-Trial.html>
34. https://www.accessdata.fda.gov/drugsatfda_docs/label/2020/761150s000lbl.pdf
35. <https://ir.zymeworks.com/News-Releases/news-details/2020/Zymeworks-Receives-FDA-Breakthrough-Therapy-Designation-for-HER2-Targeted-Bispecific-Antibody-Zanidatamab-in-Patients-with-Biliary-Tract-Cancer/default.aspx>
36. <https://investors.blackdiamondtherapeutics.com/news-releases/news-release-details/black-diamond-therapeutics-granted-fast-track-designation-fda>
37. <https://www.prnewswire.com/news-releases/remegen-announces-us-fda-has-granted-breakthrough-therapy-designation-for-disitamab-vedotin-rc48-in-urothelial-cancer-301138315.html>
38. <http://ambrx.com/fda-grants-arx788-fast-track-designation-for-her2-positive-metastatic-breast-cancer>
39. Ma et al. Neratinib Efficacy and Circulating Tumor DNA Detection of HER2 Mutations in HER2 Nonamplified Metastatic Breast Cancer. *Clin. Cancer Res.* 2017 Oct 1;23(19):5687-5695. PMID: 28679771
40. De et al. Clinical activity of afatinib (BIBW 2992) in patients with lung adenocarcinoma with mutations in the kinase domain of HER2/neu. *Lung Cancer.* 2012 Apr;76(1):123-7. PMID: 22325357
41. Kris et al. Targeting HER2 aberrations as actionable drivers in lung cancers: phase II trial of the pan-HER tyrosine kinase inhibitor dacomitinib in patients with HER2-mutant or amplified tumors. *Ann. Oncol.* 2015 Jul;26(7):1421-7. PMID: 25899785
42. Falchook et al. Non-small-cell lung cancer with HER2 exon 20 mutation: regression with dual HER2 inhibition and anti-VEGF combination treatment. *J Thorac Oncol.* 2013 Feb;8(2):e19-20. PMID: 23328556
43. David et al. Neratinib in HER2- or HER3-mutant solid tumors: SUMMIT, a global, multi-histology, open-label, phase 2 'basket' study. AACR 2017. Abstract CT001
44. Lin et al. Response to Afatinib in a Patient with Non-Small Cell Lung Cancer Harboring HER2 R896G Mutation: A Case Report. *Oncotargets Ther.* 2019;12:10897-10902. PMID: 31849493
45. Chang et al. Sustained Partial Response to Afatinib in a Patient With Lung Adenocarcinoma Harboring HER2V659E Mutation. *JCO Precis Oncol.* 2020 Aug; 912-915. DOI: 10.1200/PO.20.00114
46. Nayar et al. Acquired HER2 mutations in ER+ metastatic breast cancer confer resistance to estrogen receptor-directed therapies. *Nat. Genet.* 2019 Feb;51(2):207-216. PMID: 30531871
47. Muller et al. Mutant p53 in cancer: new functions and therapeutic opportunities. *Cancer Cell.* 2014 Mar 17;25(3):304-17. PMID: 24651012
48. Olivier et al. TP53 mutations in human cancers: origins, consequences, and clinical use. *Cold Spring Harb Perspect Biol.* 2010 Jan;2(1):a001008. PMID: 20182602
49. Guha et al. Inherited TP53 Mutations and the Li-Fraumeni Syndrome. *Cold Spring Harb Perspect Med.* 2017 Apr 3;7(4). PMID: 28270529
50. Peter et al. Comprehensive genomic characterization of squamous cell lung cancers. *Nature.* 2012 Sep 27;489(7417):519-25. PMID: 22960745
51. Cancer Genome Atlas Network. Comprehensive genomic characterization of head and neck squamous cell carcinomas. *Nature.* 2015 Jan 29;517(7536):576-82. PMID: 25631445
52. Campbell et al. Distinct patterns of somatic genome alterations in lung adenocarcinomas and squamous cell carcinomas. *Nat. Genet.* 2016 Jun;48(6):607-16. PMID: 27158780

References (continued)

53. Cancer Genome Atlas Research Network. Integrated genomic characterization of oesophageal carcinoma. *Nature*. 2017 Jan 12;541(7636):169-175. doi: 10.1038/nature20805. Epub 2017 Jan 4. PMID: 28052061
54. Olivier et al. The IARC TP53 database: new online mutation analysis and recommendations to users. *Hum. Mutat.* 2002 Jun;19(6):607-14. PMID: 12007217
55. Rivlin et al. Mutations in the p53 Tumor Suppressor Gene: Important Milestones at the Various Steps of Tumorigenesis. *Genes Cancer*. 2011 Apr;2(4):466-74. PMID: 21779514
56. Petitjean et al. TP53 mutations in human cancers: functional selection and impact on cancer prognosis and outcomes. *Oncogene*. 2007 Apr 2;26(15):2157-65. PMID: 17401424
57. Soussi et al. Recommendations for analyzing and reporting TP53 gene variants in the high-throughput sequencing era. *Hum. Mutat.* 2014 Jun;35(6):766-78. PMID: 24729566
58. <https://www.globenewswire.com/news-release/2020/10/13/2107498/0/en/PMV-Pharma-Granted-FDA-Fast-Track-Designation-of-PC14586-for-the-Treatment-of-Advanced-Cancer-Patients-that-have-Tumors-with-a-p53-Y220C-Mutation.html>
59. <https://www.aprea.com/news/aprea-therapeutics-receives-fda-fast-track-designation-and-orphan-drug-designation-for-apr-246-for-the-treatment-of-myelodysplastic-syndromes-mds/>
60. <https://ir.aprea.com/news-releases/news-release-details/aprea-therapeutics-receives-fda-fast-track-designation>
61. Parrales et al. Targeting Oncogenic Mutant p53 for Cancer Therapy. *Front Oncol.* 2015 Dec 21;5:288. doi: 10.3389/fonc.2015.00288. eCollection 2015. PMID: 26732534
62. Zhao et al. Molecularly targeted therapies for p53-mutant cancers. *Cell. Mol. Life Sci.* 2017 Nov;74(22):4171-4187. PMID: 28643165
63. NCCN Guidelines® - NCCN-Acute Myeloid Leukemia [Version 2.2021]
64. NCCN Guidelines® - NCCN-Myelodysplastic Syndromes [Version 3.2021]
65. NCCN Guidelines® - NCCN-Myeloproliferative Neoplasms [Version 1.2020]
66. NCCN Guidelines® - NCCN-Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma [Version 2.2021]
67. NCCN Guidelines® - NCCN-B-Cell Lymphomas [Version 2.2021]
68. Bernard et al. Implications of TP53 allelic state for genome stability, clinical presentation and outcomes in myelodysplastic syndromes. *Nat. Med.* 2020 Aug 3. PMID: 32747829